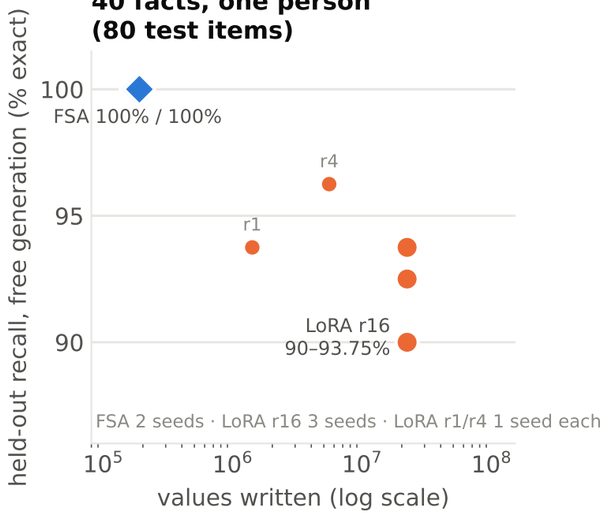
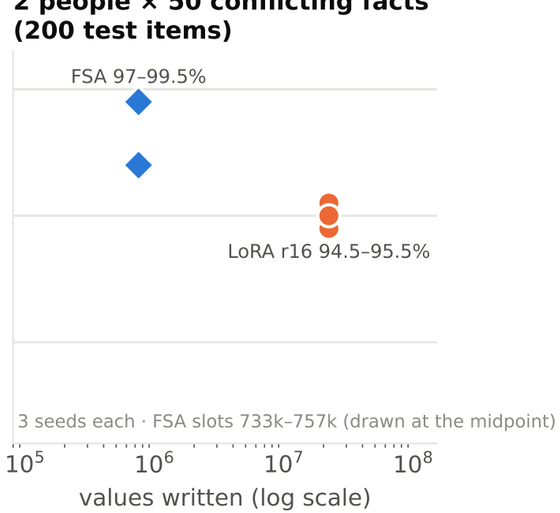
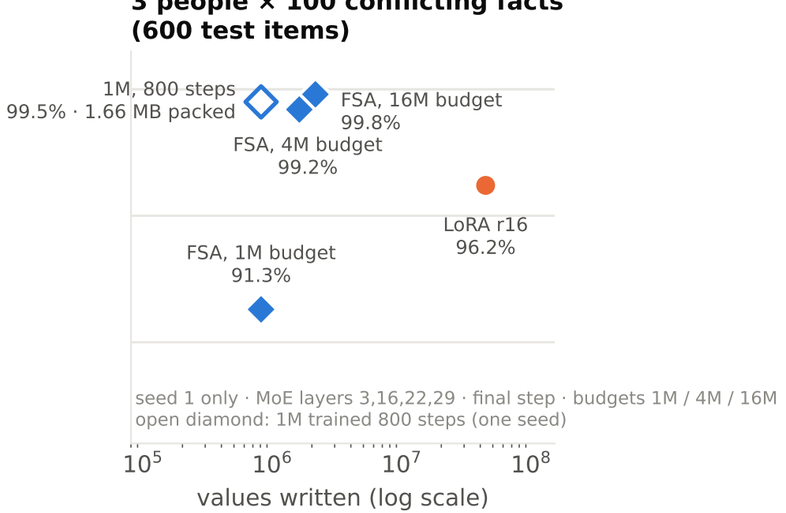
Same facts, fewer values written: the best tested FSA recipe matches or beats LoRA r16 on three fact tasks

FSA: ternary writes into the forced zeros (frozen base) LoRA adapter on the same expert matrices







3 people × 100: patch size (packed, scale derived)

	MB	stored KB/fact	vs LoRA
FSA 1M	1.66	7.15	5.5	53×
FSA 4M	2.66	8.14	8.9	33×
FSA 16M	3.31	8.80	11.0	27×
FSA 1M*	1.66	7.14	5.5	53×
LoRA r16	88.1	88.1	294	1×

FSA: measured lossless packed file, scale re-derived from the base (bit-identical on load). 1M file: 1.03 MB positions, 0.10 MB signs, 0.53 MB metadata header.
stored: same file with the per-row fp32 scale stored
LoRA: bf16 parameters
per fact = size / 300 facts
* 1M budget, 800 steps (GL_ECTRL_s1)

Test = held-out phrasings, greedy generation, whole answer must match. x: FSA = filled ternary slots, LoRA = adapter parameters (bf16 bytes = 2 × params; int4 factors would be 4× smaller). Lambda 8B ternary MoE, 725B export. 3p×100 (MoE layers 3,16,22,29): one seed each; the 400-step patches fail the +0.003 code-damage gate (FSA +0.004 / +0.011 / +0.007 at 1M / 4M / 16M, LoRA +0.008); the 1M 800-step patch (open diamond) is +0.001 ± 0.002, under the gate on the point estimate but not on its 95% upper bound, and its 400- and 800-step runs diverge from the second update for a reason not yet isolated, as well as differing in length. Recipes are tuned per method (rehearsal, penalty, steps differ), so this compares tuned recipes rather than isolating the method.